Energy communities: close the gap with local operations

Public energy-poverty data, the real cost of rural OPEX, and cooperatives as a sustainability vehicle for utilities and foundations in Colombia.



Photo: Alexander Mass / Pexels — solar on a rural home (Brazil) · Pexels licence

1 The national gap

15.4%
IMPE incidence 2024 (~8.1 M people)[1]

9.5%
2030 scenario target (~5.17 M; −2.9 M)[1]

75.9% / 93.1%
Rural vs national residential coverage (ICEE 2024)[2]

63.1% / 52.8%
IMPE La Guajira / Chocó (2024)[1]

When does the 2.9 M-person gap close?[1]



~9.7 years at observed pace · 6 years · IMPE 2030 scenario

2 OPEX, not just CAPEX

Contraloría
No methodology to pay for O&M of microgrids and PV in remote areas.[3]

Amazonas
~$30,000 M (royalties) in PV not operating; damaged batteries — abandonment risk.[4]

CREG
Thousands of rural PV systems with no management → often abandoned.[5]

CTH differentiator: we do not compete on MW funds or greenfield; we make the social and OPEX layer *bankable* where an executor and CAPEX are already committed.

[1] IMPE Report 2025 (2024 base), pp. 12, 22, 33. [2] UPME ICEE Bulletin 2024. [3] Contraloría, remote solar projects. [4] Seguimiento.co / Procuraduría, Amazonas. [5] CREG GEN-Renewables report; Concept 4022/2019 (O&M).

3 Cooperatives as the tool

CTH PROPOSAL **Community O&M + revenue + local ownership.** OPEX contracts (PV, batteries, monitoring; CREG/SSPD framework) and cooperative governance. KPIs: IMPE access/quality dimensions — not a substitute for ICEE coverage.

4 Ecosystem startups

SN
Sunco (SunColombia)
Bogotá
Design, build, and O&M for on/off-grid solar for institutions and non-interconnected zones.
898 kWp · 1,122 users (DISPAC, La Guajira)[9]
suncolombia.co

SE
SEOS Energy
Bogotá
Platform linking homes and SMEs to certified installers and solar financing.
75+ SOLvers · 18 departments[10]
seosenergy.co

SL
Soluna Energía
La Guajira · Vichada
Prepaid pay-as-you-go solar for off-grid households in the Caribbean and Orinoquía.
solunaenergia.com

TG
Tierra Grata
Cartagena
Decentralized energy and microfinance in remote rural communities (PayGo model).
1,947 clean energy solutions[11]
tierragrata.org

TW
TuWatt
Bogotá
Structures and operates distributed generation and energy-community projects with traceability.
tuwatt.co

Local talent: Escuela Colombiana de Ingeniería Julio Garavito — 101.76 kWp lab, renewable diplomas, and energy-sector challenges (Solucionatón 10X).[6]

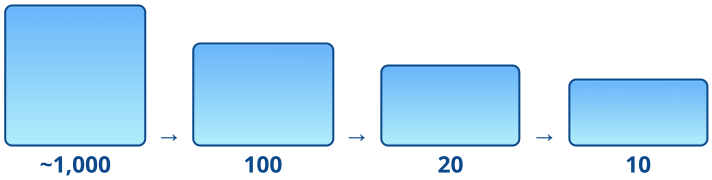
5 PASO Colombia

Productive communities in high energy-poverty departments. **PASO does not run energy projects.**[8]

- Fonseca, La Guajira (IMPE 63.1%)
- Puerto Asís / Guzmán, Putumayo (58.7%)
- Istmina, Chocó (52.8%)
- San José del Guaviare (38.9%)
- Arauca (27.9%)

6 Data lab

MinEnergía-prioritized communities → shortlist (solar resource, off-grid areas, legal status, social license).[8]



Funnel figures: illustrative.

Energy Coop candidate community: Corozal, Sucre (IMPE 43.1%) — not a PASO project.[8]

Why CleantechHUB

OPEX + cooperative
Operations contracts and local governance so CAPEX is not abandoned.

Mature projects
A *bankable* social and OPEX layer where an executor and build budget already exist — no greenfield or MW-scale funds.

IMPE KPIs + Data lab
Measurable access/quality and a candidate filter (solar, off-grid areas, legal status, social license).

[6] escuelaing.edu.co — renewables lab and continuing education. [8] CTH / PASO 2026, project brief.
[9] suncolombia.co — DISPAC project; cleantechhub.net (SUNCO portfolio profile). [10] seosenergy.co.
[11] tierragrata.org — clean energy impact.

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